

Technology News

Robot surrogate for the blind

Scientists at the California Institute of Technology have created a remote-controlled robot called CYCLOPS. It can simulate the visual experience of a blind person, implanted with a visual prosthesis such as an artificial retina.

An artificial retina consists of a silicon chip studded with a number of electrodes that directly stimulate retinal nerve cells. Scientists hope that this approach may one day help blind people to move easily and independently.

The mobile robotic platform is claimed to be the first device to emulate what a blind can see with an implant. It is in fact designed to be used as a surrogate for blind people in the testing of visual prostheses.

An artificial retina may use either an internal or external miniature camera to capture images. These images are then processed and passed on to the implanted silicon chip's electrode array. The chip directly stimulates the eye's functional retinal ganglion cells that carry the image information to

the vision centres in the brain.

If scientists want to see how much better the resolution is when a retinal prosthesis has an array of 50 pixels as opposed to 16 pixels, they can try both on CYCLOPS. They can do this by asking the robot to follow a black line down a white-tiled hallway or see if it can find and enter a darkened doorway.

Penny-sized nuclear battery

Batteries can power anything, from small sensors to large systems. While scientists are finding ways to make them smaller yet more powerful, problems can arise when these batteries are larger and heavier than the devices. Researchers at the University of Missouri are developing a nuclear energy battery that is smaller, lighter and more efficient.

The research team has been working to make the nuclear battery smaller like the size and thickness of a penny. It is intended to power various micro/nano-electromechanical systems. The research team claims that although nuclear batteries can pose concerns, this battery is safe.

It uses a liquid semiconductor and not a usual solid one.

World's smallest watch phone

Here comes the world's smallest wristwatch cellphone which sports a calendar, address book, voice mail, SMS, call log, speed dial, calculator, stopwatch, a vibration mode for silent



Kempler & Strauss W PhoneWatch

alarms, a no-stylus touchscreen, microSD slot and the ability to take and play pictures and videos.

San Diego-based Kempler & Strauss has launched this GSM quad-band cellphone watch called W PhoneWatch. The phone, which appears like an attractive digital wristwatch, is a full-function integrated advanced communications device. The W PhoneWatch incorporates a GSM phone that supports 850MHz, 900MHz, 1800MHz and 1900MHz bands. The TFT LCD micro-touch touchscreen displays 260,000 colours. The watch measures 57 x 45 x 18 mm and weighs 71 gms.

Fully compatible with Bluetooth v2.0 headset and hands-free devices, the W PhoneWatch incorporates a still camera that captures images at resolutions up to 640 x 480 pixels. It also captures video in MPEG 4 (AVI) format and supports video playback of MPEG4 files and music files including MP3, WMA, WAV, and AMR.

Move to make cells less expensive

Fuel cells that can produce electricity from hydrogen or other fuels without burning them are considered



The CYCLOPS mobile robotic platform